is to stand ready to deliver such products on a when-and-if-available basis.

Software support revenue is typically recognized based on time elapsed and thus ratably over the term of the support arrangement. Under our standardized support services, our performance obligation is to stand ready to provide technical product support and unspecified updates, upgrades, and enhancements on a when-and-if-available basis. Our customers simultaneously receive and consume the benefits of these support services as we perform.

Service revenue is typically recognized over time. Where we stand ready to provide the service (such as access to learning content), we recognize revenue based on time elapsed and thus ratably over the service period. Consumption-based services (such as separately identifiable consulting services and premium support services, messaging services, and classroom training services) are recognized over time as the services are utilized, typically following the percentage-of-completion method or ratably. When using the percentage-of-completion method, we typically measure the progress toward complete satisfaction of the performance obligation in the same way and with the same reasoning and judgment as we do for customer-specific on-premise software development agreements. We apply judgment in determining whether a service qualifies as a stand-ready service or as a consumption-based service.

Revenue for combined performance obligations is recognized over the longest period of all promises in the combined performance obligation.

Judgment is also required to determine whether revenue is to be recognized at a point in time or over time. For performance obligations satisfied over time, we need to measure progress using the method that best reflects SAP's performance. When using cost incurred as a measure of progress for recognizing revenue over time, we apply judgment in estimating the total cost to satisfy the performance obligation.

All of the judgments and estimates mentioned above can significantly impact the timing and amount of revenue to be recognized.

Contract Balances

We recognize trade receivables for performance obligations satisfied over time gradually as the performance obligation is satisfied and in full once the invoice is due. Judgment is required in determining whether a right to consideration is unconditional and thus qualifies as a receivable.

Contract liabilities primarily reflect invoices due or payments received in advance of revenue recognition.

Typically, we invoice fees for on-premise standard software on contract closure and software delivery. Periodic fixed fees for cloud subscription services, software support services, and other multi-period agreements are typically invoiced yearly or quarterly in advance. Such fee prepayments account for the majority of our contract liability balance. Fees based on actual transaction volumes for cloud subscriptions and fees charged for non-periodical services are invoiced as the services are delivered. While payment terms and conditions vary by contract type and region, our terms typically require payment within 30 to 60 days.

Geographic Information

The amounts for revenue by region in the following tables are based on the location of customers. The regions in the following table are EMEA (Europe, Middle East, and Africa), Americas (North America and Latin America), and APJ (Asia Pacific Japan).

Total Revenue by Region

€ millions	**2019**	2018	2017
Germany	3,948	3,658	3,352
Rest of EMEA	8,158	7,446	7,063
EMEA	**12,105**	**11,104**	**10,415**
United States	9,085	7,880	7,436
Rest of Americas	2,109	1,832	1,911
Americas	**11,194**	**9,713**	**9,347**
Japan	1,180	963	885
Rest of APJ	3,074	2,928	2,814
APJ	**4,254**	**3,891**	**3,699**
SAP Group	**27,553**	**24,708**	**23,461**

Major Revenue Classes by Region

€ millions		Cloud Revenue			Cloud and Software Revenue	
	2019	2018	2017	**2019**	2018	2017
EMEA	2,115	1,441	1,029	10,211	9,339	8,759
Americas	3,945	2,941	2,321	9,172	7,973	7,666
APJ	872	611	419	3,629	3,310	3,124
SAP Group	**6,933**	**4,993**	**3,769**	**23,012**	**20,622**	**19,549**

For information about the breakdown of revenue by segment and segment revenue by region, see *Note (C.1)*.